Rotations

Exam code: 0607

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Written by: Dan Finlay
Reviewed by: Lucy Kirkham
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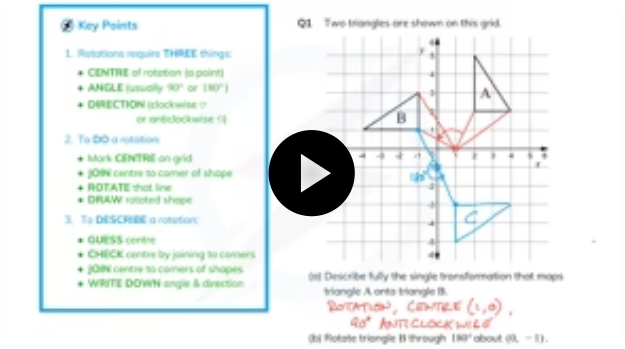

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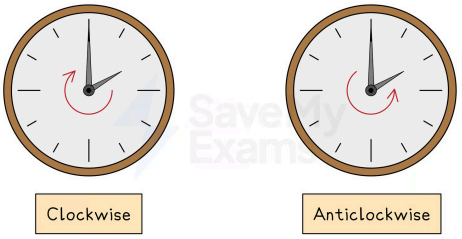
Rotations

What is a rotation?

- A **rotation turns** a shape **around a point**
  - This is called the **centre of rotation**
- The **rotated image** is the **same size** as the **original image**
  - It will have a **new position and orientation**
- If the centre is a point on the original shape then that point is **not changed** by the rotation
  - It is called an **invariant point**



Orientation of a rotation

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I rotate a shape?

- **STEP 1**

Place the point of your pencil on the **centre of rotation**

- **STEP 3**
  Rotate the tracing paper by the **given angle** in the **given direction**
  - The angle will be **90°**, **180°** or **270°**

- **STEP 4**
  Carefully draw the image onto the coordinate grid in the **position shown by the tracing paper**

Examiner Tips and Tricks

When you first go into the exam room, make sure there is some **tracing paper** on your desk ready for you. If there isn't ask for some before the exam begins.

Draw an **arrow facing up** on your tracing paper.

The arrow will be **facing left or right** when you have turned **90° or 270°.**

The arrow will be **facing down** when you have turned **180°.**

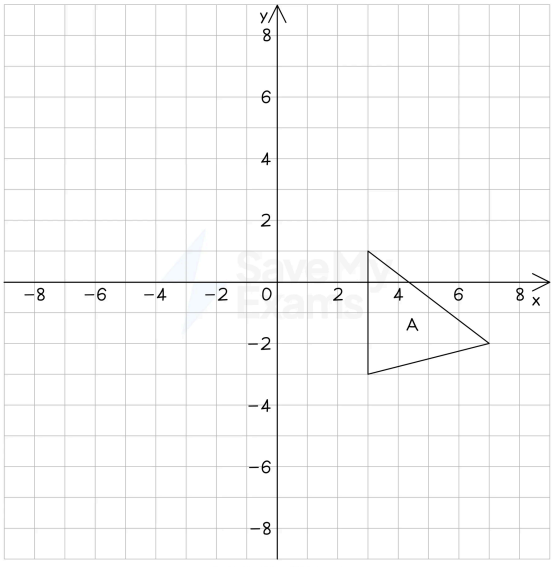
Double-check that you have copied the rotated image into the correct position.

Put the tracing paper over the original object and rotate it again to see that it lines up with your image.

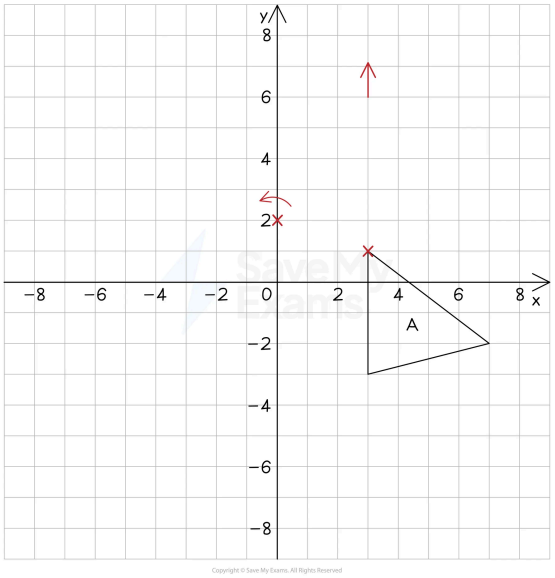
Worked Example

(a) On the grid below rotate shape A by 90° anti-clockwise about the point (0, 2).

Label your answer A'.



one vertex.
Mark on the centre of rotation.
Draw an arrow pointing vertically upwards on the paper.



With your pencil fixed on the point of rotation, rotate the tracing paper 90° anti-clockwise, the arrow that you drew should now be pointing left.
Make a mental note of the new coordinates of the vertex that you marked on your tracing paper.
Draw the new position of this vertex onto the grid.

Repeat this process for the other two vertices on the triangle.
Connect the vertices together to draw the rotated image.

How do I describe a rotation?

- To describe a **rotation**, you must:
  - State that the transformation is a **rotation**
  - State the **centre of rotation**
  - State the **angle of rotation**
    - This will be 90°, 180° or 270°
  - State the **direction of rotation**
    - Clockwise or anti-clockwise
    - A direction is not required if the angle is 180°
    - 90° clockwise is the same as 270° anti-clockwise
- To find the **centre of rotation**:
  - If the rotation is **90° or 270°**
    - Use **tracing paper** and **start on the original shape**
    - **Try a point** as the centre and rotate the original shape
    - If the **rotated shape matches the image** then that point is the centre
    - Otherwise keep picking points until one works
  - If the rotation is **180°**
    - Draw **lines connecting** each vertex on the **original shape** with the corresponding vertices on the **image**
    - These lines will **intersect** at the centre of rotation

How do I reverse a rotation?

- If a shape has been **rotated** to a new position, you can perform a single transformation to **return the shape** to its **original position**
- A rotation can be **reversed** by simply **reversing the direction** of rotation
  - The angle of rotation is the same
  - The centre of rotation is the same
- For a shape rotated by 45° in a clockwise direction about the point (0, 3)
  - The **reverse transformation** is
    - a rotation of 45°

Worked Example

Describe fully the single transformation that creates shape B from shape A.

Answer:

You should be able to see that the object has been rotated 90° clockwise (or 270° anti-clockwise).
You are likely to be able to see roughly where the centre of rotation is but it may take a little time to find its position exactly.

To find the exact coordinates of the centre of rotation you can play around with tracing paper.

seeing if it lines up with shape B.

Write down the all of the elements required to fully describe the transformation: the type of transformation, the centre of rotation, the angle and the direction.

Rotation, 90° clockwise with centre (−4, 0)

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